



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)
Integrated Bachelor's-Master's Program
Mid-Semester Examination (Sem-I) - Autumn 2025

Subject: Mathematics I

Full marks: 25

Subject Code: MAT 1101

Time allotted: 2 hours

- Please READ the following INSTRUCTIONS carefully before starting.
 - You MUST ATTEMPT Question 1, which carries 5 marks.
 - Questions 2-5 carry 27 marks. You may attempt AS MUCH AS YOU CAN, but the MAXIMUM you can score in Questions 2-5 is 20 marks.
1. For each of the following statements, simply say whether it is **TRUE** or **FALSE**.
No explanation/justification is necessary! (5 × 1) = [5]
- (a) $(A \cup B) \cap (B \cup C) \cap (A \cup C) = (A \cap B) \cup (B \cap C) \cap (A \cap C)$.
 - (b) If $f : A \rightarrow B$, then $f(C \cap D) = f(C) \cap f(D)$ for any two subsets $C, D \subset A$.
 - (c) If $f : A \rightarrow B$, $g : B \rightarrow A$ satisfy $g \circ f(x) = x$ for all $x \in A$, then f is one-one.
 - (d) If $A \cup B \sim \mathbb{N}$, then at least one of $A \sim \mathbb{N}$ or $B \sim \mathbb{N}$ must hold.
 - (e) If A, B and C are sets such that $A \times B \sim A \times C$, then $B \sim C$ must hold.
2. Prove that, for all $n \in \mathbb{N}$, the number $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9. [6]
3. Fix a natural number $n > 1$. Show that, if $f : \mathbb{N}_n \rightarrow \mathbb{N}_n$ is any bijection with $f(1) = n$, then $g(k) = f(k+1)$, $k \in \mathbb{N}_{n-1}$, defines a bijection on $\mathbb{N}_{n-1}$ onto itself. Conversely, given any bijection $g : \mathbb{N}_{n-1} \rightarrow \mathbb{N}_{n-1}$, show that $f : \mathbb{N}_n \rightarrow \mathbb{N}_n$, defined as $f(1) = n$ and $f(k) = g(k-1)$, $k > 1$, is a bijection. [7]
4. Show that if $f : A \rightarrow B$ and $g : B \rightarrow C$ are such that $g \circ f$ is surjective, then g must be surjective. Give an example to show that the converse may not hold. [7]
5. Prove that, for each $n \in \mathbb{N}$, the set of all polynomials of degree n with integer coefficients is a countable set. Deduce that the set of all polynomials with integer coefficients is a countable set. [7]